

CSR and sustainability reporting practices in India: an in-depth content analysis of top-listed companies

Priyanka Aggarwal and Ajay Kumar Singh

Abstract

Purpose – *The purpose of this paper is to comprehensively analyze the corporate social responsibility (CSR) and sustainability reporting (SR) practices of Indian companies in terms of disclosure quantity and quality, and to investigate the differences in SR practices by SR dimension, industry, ownership structure, firm size and profitability.*

Design/methodology/approach – *Data are collected from annual reports/business responsibility reports (BRR)/CSR/sustainability reports of 60 top-listed companies in India. A comprehensive sustainability reporting index is developed. Content analysis technique is used. Inter-coder reliability is established.*

Findings – *Altogether, 18 items of the index are not disclosed by the majority of companies in India. SR quality is found significantly lower than the SR quantity. Moreover, SR practices significantly differ by dimension/category, industry-type and firm-size but are not influenced by ownership structure. However, the study fails to establish any conclusive relationship between SR and profitability.*

Practical implications – *The present study has several implications for corporates, practitioners, policymakers and stakeholders. The findings underscore the need for amendments in the Global Reporting Initiative guidelines and BRR framework of the Securities and Exchange Board of India to avoid patchy disclosures and ensure complete reporting by companies.*

Originality/value – *This study is among the foremost studies in India evaluating SR practices of top-listed companies in the wake of the mandatory BRR requirement from a quantitative as well as qualitative perspective using a multidimensional index.*

Keywords *Content analysis, Integrated reporting, Sustainability reporting, Corporate social responsibility (CSR), Global Reporting Initiative (GRI), Business responsibility reporting (BRR)*

Paper type *Research paper*

Priyanka Aggarwal is based at the Shri Ram College of Commerce, University of Delhi, New Delhi, India. Ajay Kumar Singh is based at the Department of Commerce, Delhi School of Economics, University of Delhi, New Delhi, India.

Introduction

The concept of corporate social responsibility (CSR) dates back to the 1950s. Howard R. Bowen – considered the pioneer of CSR – defined CSR as “the obligations of businessmen to pursue those policies, to make those decisions, or to follow those lines of action which are desirable in terms of objectives and values of our society” (Bowen, 1953, p. 6).

The stakeholders, being more cognizant, have been progressively demanding information on non-financial performance of companies. Sustainability reporting (SR) is, thus, a pertinent issue for companies today. It is defined by the Global Reporting Initiative (GRI) as –“the practice of measuring, disclosing and being accountable to internal and external stakeholders for organizational performance toward the goal of sustainable development” (Global Reporting Initiative, 2011).

Received 27 March 2018
Revised 3 May 2018
14 June 2018
9 August 2018
Accepted 29 September 2018

Note: There is no potential conflict of interest in this research.

According to KPMG, “sustainability reports include quantitative and qualitative information on financial, social and environmental performance of companies in a balanced way” (Kolk *et al.*, 2002).

According to the KPMG Survey 2015, four developing countries including India emerged as frontrunners in SR; however, Europe was the leader in 2011 (KPMG, 2015). The quality and reliability of CSR/SR is a great concern even for developed nations; e.g. Swiss companies have an average SR score of only 33 per cent (Daub, 2007). Moreover, multiple standards exist in this realm, e.g. GRI, SA8000, ISO 26000, etc. causing inconsistency in SR. India is among the leading countries that formally recognizes CSR with mandatory CSR spending and reporting regulations. The purpose of this paper is to assess the status of CSR/SR in India from the quantitative and qualitative perspectives, especially in the wake of mandatory BRR requirement by the SEBI. Particularly, the present study seeks to explore the following research questions (RQs) in Indian context:

- RQ1. Which *sustainability issues/topics* currently lack attention from Indian corporates and managers?
- RQ2. What are the plausible motives behind sustainability reporting in India – regulatory pressure vs voluntary approach?
- RQ3. Is there any significant difference between the quantity and quality of sustainability disclosures?
- RQ4. Do SR practices differ across dimensions/categories?
- RQ5. Do SR practices differ by *industry-type*?
- RQ6. Do SR practices of *environmentally sensitive* companies differ from other companies?
- RQ7. Is there any difference in SR by *ownership structure*?
- RQ8. Is SR influenced by economic/corporate financial performance (*CFP*) *indicators*?

Theoretical background

The drivers of SR include firm-size, profitability, age, media-exposure, ownership structure and capital structure (Dienes *et al.*, 2016). The probable motives of companies behind undertaking SR are discussed below.

Regulatory pressure

In some form, CSR/SR has entered into the realm of mandatory requirements in several countries. The G20 members – European Union and 19 countries including India, China and the UK and USA– require environmental, social and governance (ESG) disclosures. Such regulations impel companies to undertake SR to avoid disciplinary actions. This can be linked to the *legal dimension of CSR*, wherein organizations are expected to meet only *minimum legal requirements* (Carroll, 1991). Thus, if a company makes meager disclosures only to comply with the SEBI BRR requirement and doesn't voluntarily disclose information as per global standards such as GRI; such a company is perhaps reporting merely out of regulatory pressure.

Legitimacy theory

Legitimacy theory proposes existence of an apparent contract between the company and society, which compels companies to meet societal norms to ensure their survival (Burhan and Rahmanti, 2012; Schmelzer, 2013). CSR/SR lends legitimacy/acceptability to corporate actions by influencing the public perception affirmatively and helps in avoiding undesirable media coverage (Patten, 1992; Deegan, 2002; Campbell *et al.*, 2003). Thus, legitimacy

theory propagates the *ethical dimension of CSR*, wherein firms are expected to comply with social norms beyond legal requirements (Carroll, 1991). Thus, if a company makes considerably more society/community disclosures over other dimensions, it is possibly driven by legitimacy theory.

Stakeholder theory

Stakeholder theory advocates that organizations should pursue the interests of all stakeholder groups, including investors, customers, suppliers, employees, government, community and environment in an equitable manner (Clarkson, 1995). Stakeholders are capable of affecting the organization's performance and reputation. SR helps firms in building strong ties with stakeholders yielding several benefits such as lower risk, enhanced reputation and competitive advantage (Barnett, 2007; Schmelzer, 2013). Stakeholder theory corresponds to the *discretionary dimension of CSR*, which urges organizations to be good corporate citizens who contribute back to the society while going beyond social norms and laws (Carroll, 1991). Therefore, the companies that voluntarily disclose ESG information with quantitative data while catering to stakeholders across-the-board are adopting the stakeholder theory perspective.

Literature review and hypotheses development

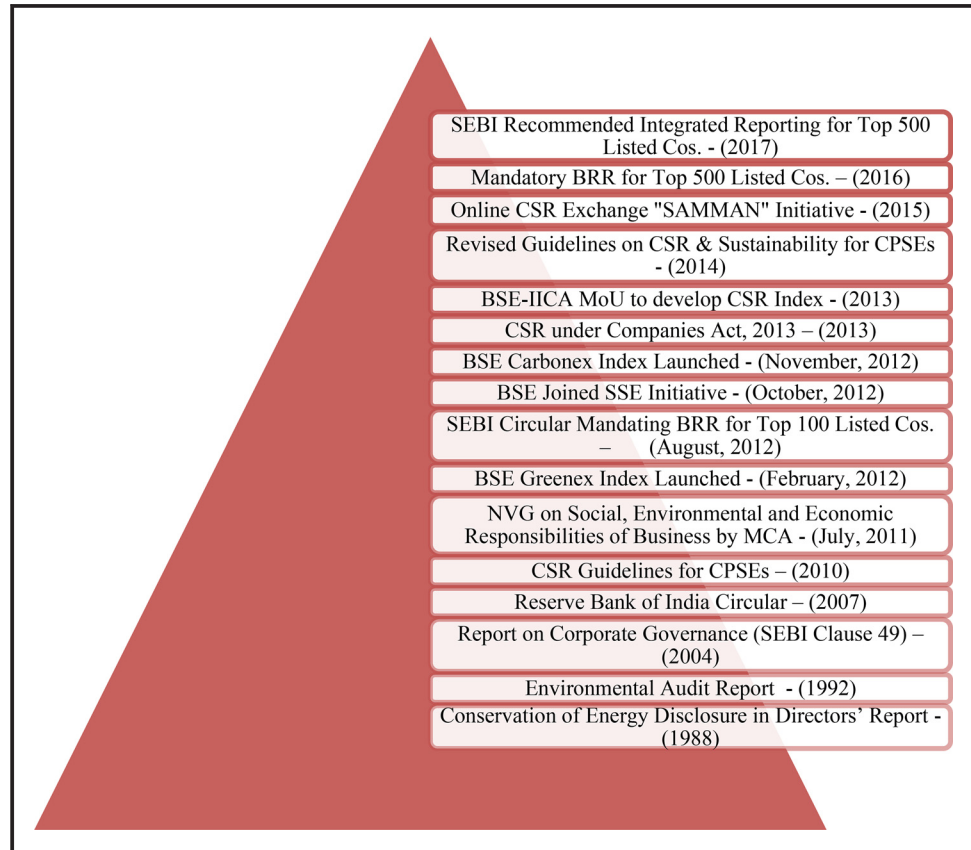
CSR research gained popularity in the 1980s. Initially, CSR studies were conducted only in developed countries such as the USA and UK. Several previous researchers have investigated the determinants of CSR/sustainability disclosures in terms of firm-specific factors such as firm-size, profitability, industry and ownership structure (Jain and Winner, 2016; Mertens et al., 2012).

Corporate social responsibility/sustainability reporting in India

CSR has become a buzzword for India Inc. in recent years, particularly after the enactment of the New Companies Act, 2013, which requires companies to spend at least 2 per cent of their average profits on CSR activities. CSR/SR in India commenced in the 1980s with the mandatory disclosure on the Conservation of Energy in Directors' Report. Over the years, India has progressed radically in the sphere of CSR. In 2012, the SEBI made it mandatory for the top 100 listed companies to practice the National Voluntary Guidelines (NVG) on social, environmental and economic responsibility issued by the Ministry of Corporate Affairs (MCA) and to uniformly disclose their CSR activities in BRR as part of the Annual Report (SEBI, 2012, August 13). From April 1, 2016 onward, the BRR requirement has been extended to the top 500 listed companies. SEBI has also clarified that failure to provide BRR shall be interpreted as non-compliance with Clause 55 of the listing agreement calling for grave aftermaths such as heavy penalties, fines, suspensions and even delisting if non-compliance persists. Recently, the SEBI recommended and urged the top 500 listed companies to embrace integrated reporting on a voluntary basis from financial year (FY) 2017-18. Figure 1 illustrates the progression of CSR/sustainability in India.

Only a few studies have investigated CSR/SR practices in India, especially after the aforesaid regulatory developments. Some notable Indian studies in recent times are Laskar and Maji (2016); and Jain and Winner (2016). The previous studies largely analyzed CSR disclosures in terms of presence/absence in corporate reports while ignoring the quality aspect. Also, social and environmental dimensions alone have been emphasized hitherto, while there is a dearth of research on other dimensions such as human rights and governance (Murray et al., 2006; Bayoud et al., 2012; Khavesh et al., 2012). Moreover, previous studies predominantly assessed disclosures based on the GRI guidelines and overlooked the Indian scenario. The current study seeks to bridge this gap.

Figure 1 Progression of CSR and sustainability in India



Sustainability reporting quantity and quality

Previous studies have mostly used binary scale (0-1; [Yang et al., 2010](#); [Kapoor and Sandhu, 2010](#); [Faisal et al., 2012](#)), while only a few studies used multiple-scoring scale to evaluate CSR disclosures ([Yi and Yu, 2010](#); [Guidry and Patten, 2010](#); [Ameer and Othman, 2012](#)). On the basis of graphical analysis, [Laskar and Maji \(2016\)](#) concluded that a difference between level (0-1) and quality (four-point scale) of CSR disclosures is insignificant and that both have the same impact on CFP. There is a need to empirically probe whether the number of sustainability disclosures (quantity) and quality thereof are essentially different. Hence, the first hypothesis is as follows:

H1. There is a significant difference between SR quantity and SR quality scores of companies.

Sustainability reporting along different sub-categories/dimensions

Sustainability reporting is a multidimensional concept. In this study, we have analyzed SR practices along with seven dimensions, namely, economic, governance and ethics, environment, community, customers, employees and human rights. [Jain and Winner \(2016\)](#) found the environment to be the most addressed CSR dimension, followed by health and education. The current study explores how SR practices differ among various dimensions. Thus, the second hypothesis is as follows:

H2. There is a significant difference in sustainability reporting by dimension.

H2a. There is a significant difference in SR quantity by dimension.

H2b. There is a significant difference in SR quality by dimension.

Sustainability reporting and industry-type

Industry is another important variable that influences SR as different industries are subject to diverse social–environmental concerns. The companies belonging to environmentally sensitive industries (ESI) such as oil and gas and metals/mining are generally subjected to more stakeholder pressure than others (Neu *et al.*, 1998; Guidry and Patten, 2010; Reverte, 2012). Stakeholders, especially investors, pay more attention toward CSR performance of ESI companies, demanding more CSR information (Choi, 1999; Frost and Wilmshurst, 2000; Young and Marais, 2012). Consequently, ESI companies make more CSR disclosures to improve stakeholders' perceptions (Patten, 1992; Pled and Latridis, 2012). Moreover, a few studies have concluded that there exists a more evident positive association between CSR disclosure quality and CFP of ESI firms (Plumlee *et al.*, 2008; Reverte, 2012). Thus, it is critical to investigate the role of industry-type in general and ESI, in particular, in influencing SR practices in India. Consequently, the third and fourth hypotheses are as follows:

H3. Sustainability reporting (SR) practices differ by industry type.

H3a. SR quantity differs by industry type.

H3b. SR quality differs by industry type.

H4. SR practices of ESI companies are significantly different from non-ESI companies.

H4a. SR quantity of ESI companies is significantly different from non-ESI companies.

H4b. SR quality of ESI companies is significantly different from non-ESI companies.

Sustainability reporting and ownership structure

The study further attempts to gain deeper insights into SR practices in India on the basis of ownership structure, i.e. public *vis-à-vis* private sectors. Public sector enterprises are those in which Government has a major shareholding, while private sector companies have no Government stake. There are contrasting views about the importance of CSR in public and private sector organizations. One school of thought upholds that private sector companies are allegedly less CSR-oriented as they are highly profit-oriented, unlike public-sector undertakings (PSUs), which are governed by social objectives. Jain and Winner (2016) also concluded that public-sector companies make better CSR disclosures than private companies. In contrast, Sahu and Panigrahy (2016) observed that companies in the private sector perform better in CSR. To review this issue, the following fifth hypothesis is developed:

H5. SR practices differ by ownership structure.

H5a. SR quantity differs by ownership structure.

H5b. SR quality differs by ownership structure.

Sustainability reporting and economic performance

Several researchers in past have examined the linkage between SR and CFP (Buys *et al.*, 2011; Burhan and Rahmanti, 2012). According to *slack resources view*, financially prosperous companies have more surplus resources and hence expected to be more committed toward CSR (Udayasankar, 2008; Palmer, 2012). According to *political cost perspective*, big companies are more exposed to media, public and socio-regulatory pressure and hence are more likely to communicate their CSR activities to gain legitimacy (Cormier and Magnan, 2003; Setyorini and Ishak, 2012). Drawing on aforesaid arguments,

CSR/SR has been examined *vis-à-vis* two economic performance indicators – firm-size and profitability. Consequently, the sixth and seventh hypotheses are as follows:

H6. Large-sized companies perform better in terms of sustainability reporting.

H6a. Large-sized companies have higher SR quantity.

H6b. Large-sized companies have better SR quality.

H7. More profitable companies perform better in terms of SR.

H7a. More profitable companies have higher SR quantity.

H7b. More profitable companies have better SR quality.

Research design: sample, data and method

Sample selection

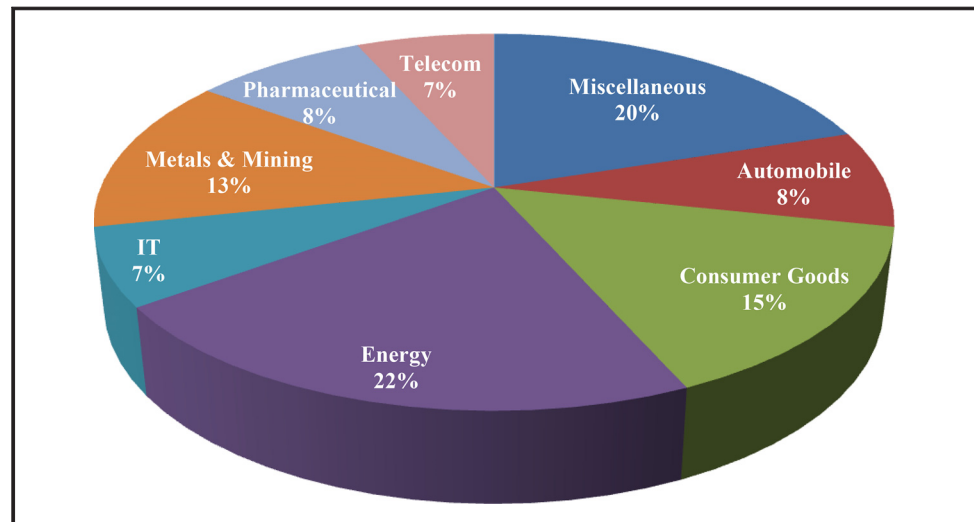
The final sample consists of 60 non-financial companies having BRR/CSR/SR selected out of top 100 companies listed at NSE (India). Sample distribution by industry type is described in [Figure 2](#).

Data

CSR/sustainability data of selected companies for the year ended March 31, 2014 have been exhaustively searched and obtained from corporate websites, CSR/sustainability/BRR/Annual Reports, CSR Policy Statements, etc. 2013-2014 has been chosen as study period since India witnessed major CSR reforms during 2013, e.g. formal recognition of CSR under the New Companies Act, 2013 and regulations on mandatory CSR spending and BRR.

Further, financial data are retrieved from the Prowess database. The study uses two economic/CFP indicators – firm-size and profitability. On the basis of past studies, firm size is measured using natural logarithm of market capitalization, while profitability is measured using return on assets (ROA; [Manescu, 2011](#); [Moneva and Cuellar, 2009](#)).

Figure 2 Sample distribution by industry-type



Content analysis process

Content analysis is a qualitative research technique used to interpret and draw inferences in an objective/systematic and quantifiable manner by evaluating textual material such as reports against predetermined criteria. It has been extensively used in previous CSR communication research (Abbott and Monsen, 1979; Morhardt et al., 2002; Daub, 2007; Vormedal and Ruud, 2009).

First, we developed a comprehensive Sustainability Reporting Index (SRI) based on several standards and guidelines, particularly bearing in mind the Indian scenario, viz. GRI G4 guidelines, UNGC Principles, ISO 26000, SA 8000, NVG by MCA, SEBI-BRR Framework and Companies Act, 2013. The 80-item equally weighted index comprises 7 dimensions, namely, economic (8 items), governance and ethics (15), environment (16), community (9), customers and product responsibility (8), employees and labor practices (17) and human rights (7), as presented in Figure 3. See “Appendix 1” for the complete index.

Subsequently, coding has been done by authors from a qualitative and quantitative perspective. To examine SR quantity, the presence of each item is checked in the corporate reports and website of each company and a score of 0 (absence) or 1 (presence) is assigned accordingly. The disclosures are further assessed in terms of quality using a six-point (0-5) scale, which has been used for the first time for evaluating CSR/sustainability disclosures. The scale has been developed based on previous studies – Daub (2007), Vormedal and Ruud (2009) – and is described in Table I. The scores so obtained are then expressed in percentage, wherein higher scores reflect the better ability of the company to communicate with stakeholders transparently.

Besides, a third-party/independent researcher also performed the content analysis of reports for a sub-sample representing around 10 per cent of the whole sample. Inter-coder reliability (ICR) estimation is critical for establishing the trustworthiness of data generated

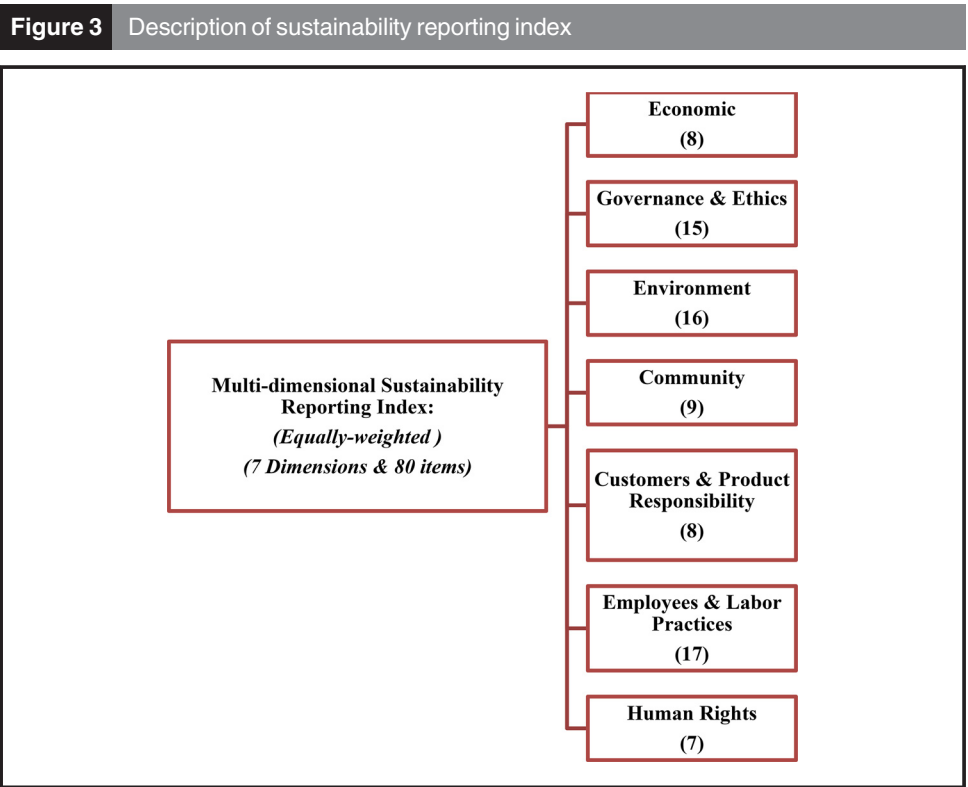


Table I Sustainability reporting quality assessment scale

Score	Criterion	SR quality
0	No disclosure	–
1	Minimal, general and non-quantitative disclosure	Poor
2	Descriptive and company-specific/quantitative disclosure, but <i>many</i> relevant points not addressed	Below average
3	Descriptive and company-specific/quantitative disclosure, but <i>some</i> relevant points not addressed	Average
4	Descriptive and company-specific/quantitative Disclosure, but <i>few</i> relevant points not addressed	Above average
5	Descriptive and company-specific/quantitative disclosure, with <i>all</i> relevant points addressed	Excellent

through content analysis. ICR was tested using intra-class correlation coefficient and Krippendorff's alpha test (Hayes and Krippendorff, 2007) and was found to be 0.956 and 0.984, respectively, indicating excellent inter-coder agreement and hence, our SR scores can be considered reliable.

Data analysis techniques

A variety of quantitative techniques were used to test the research hypotheses. Descriptive statistics (e.g. mean and standard deviation) were computed. The differences in SR practices industry-type, firm-size, etc. were examined through *t*-test and ANOVA. Further, the effect size was measured using eta-squared and Cohen's *d* and *post-hoc* Tukey's test. Also, the association between SR and CFP was analyzed using Pearson's correlation.

Results and findings

Sustainability reporting practices in India

Results revealed that approximately one-third (36.67 per cent) sample companies publish standalone CSR/SR while a mere 28.33 per cent companies follow GRI guidelines. Around one-fourth companies issue both BRR and GRI reports, and only 23.33 per cent companies publish externally assured GRI report. These statistics demonstrate sparseness of sustainability reporting in India. As the adoption of the GRI framework is limited even among the top-notch companies, regulatory pressure seems to be the major motive behind sustainability reporting in India.

Wipro Ltd. is the only company to have voluntarily sought external-assurance of BRR. Besides, all companies have prepared BRR, except Marico Industries Ltd. and Coal India Ltd.; however, they have published a separate sustainability report. SEBI allows companies not to prepare BRR if BRR principles are mapped with sustainability disclosures. Further analysis reveals that Coal India Ltd. published separate GRI-based sustainability report with mapping, however, Marico had not followed any international standard; moreover, mapping was omitted. Thus, Marico, a renowned household brand, happened to be a defaulter for FY 2013-2014. Nevertheless, Marico soon mended the reporting pattern, and the subsequent sustainability reports of Marico were in compliance with GRI and SEBI guidelines. Besides, a lack of uniformity is observed in the sustainability reporting formats adopted used in India, which deters understandability and comparability.

Top three companies based on SR quantity and quality scores are shown in Table II. GAIL is adjudged as leading company in SR quality with a 79.25-per-cent score, while Tata Power Ltd. has the highest SR quantity score having reported 95 per cent items. Further, Tata Steel Ltd. has outrivalled both in SR quantity and quality. It is worth noting that all leading companies belong to ESI. Thus, ESI companies certainly perform better in SR, perhaps to improve their public image. However, companies with the *lowest SR quality* scores are Sun TV Network Ltd., Zee Entertainment Ltd. and Reliance Power Ltd. Moreover, Sun TV and Reliance Power have the lowest SR quantity scores as well, besides Marico

Table II Leading sustainability reporting companies

Company	SR quality score (%)	SR quantity score (%)	Rank as per SR quality	Rank as per SR quantity
Top three companies, as per SR quality				
GAIL	79.25	92.50	1	5
Cairn	73.50	86.25	2	14
Tata Steel	73.25	93.75	3	2
Tata Motors	73.25	92.50	3	5
Top three companies as per SR quantity				
Tata Power	73.00	95.00	5	1
Tata Steel	73.25	93.75	3	2
Indian Oil Corporation	70.25	93.75	8	2

Industries Ltd. Ranks and SR scores of all 60 sample companies are presented in “Appendix 2”.

Further, it is discovered that altogether 18 index items are not disclosed by the majority of companies. Predominantly, a dearth of disclosures on the following topics has been identified: local hiring procedure, risk assessment against corruption/unethical practices, performance evaluation of governing-body, pay-gap, external-assurance, impact on biodiversity, direct/indirect emissions, environment protection expenditure, political contributions, sale of banned/disputed/hazardous products, exclusive benefits for full-time employees, minimum-wages for employees, minimum notice period about operational changes, work-related fatalities rate and employee training on human rights. These issues demand special attention from corporates and managers to uplift the level of SR in India.

Quantity and quality of sustainability reporting

First, descriptive statistics, as reported in Table III, indicate that average reporting is done on three-fourth items that too of mediocre quality. Further, the majority of companies have “Below Average” SR quality, 25 per cent have “Above Average” quality, and none of the companies has excellent quality, as is shown in Figure 4.

Further, it is found that there is a significant difference between the SR quality and quantity scores ($p = 0.00$). Moreover, the *effect size is large*, exceeding 0.8 (Cohen's $d = 1.9$), implying that SR quality is substantially lower than SR quantity, providing *support for H1*. However, the difference between rankings on the basis of the aforesaid scores is not statistically significant ($p = 0.88$), as can be seen in Table III. This is probably owing to a high positive correlation between SR quality and SR quantity ($r = 0.941$; $p = 0.00$)

Sustainability reporting differences by sub-categories/dimensions

Among the seven SR dimensions, *Community disclosures are the best*, while *employee disclosures are the poorest* both in quantity/frequency and quality, as is shown in Figure 5.

Table III Descriptive statistics

Particulars	SR_quality (%)	SR_quantity (%)
Mean	48.50	74.77
Median	45	73.125
SD	14.99	12.435
Minimum	19.24	48.10
Maximum	79.25	95

Figure 4 Frequency distribution *vis-À-vis* sustainability reporting quality

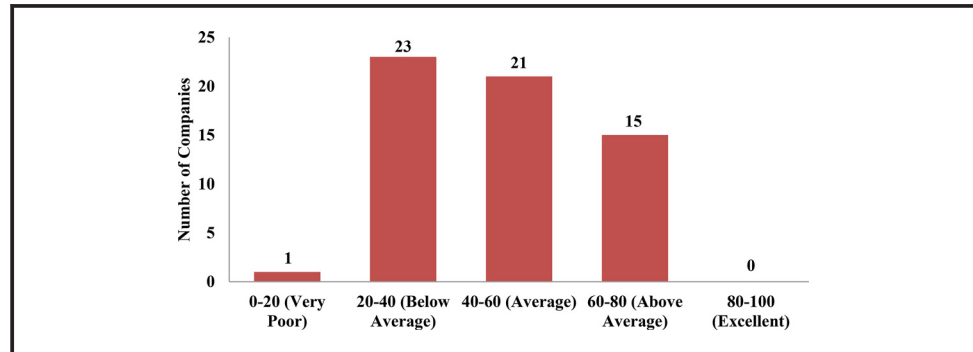
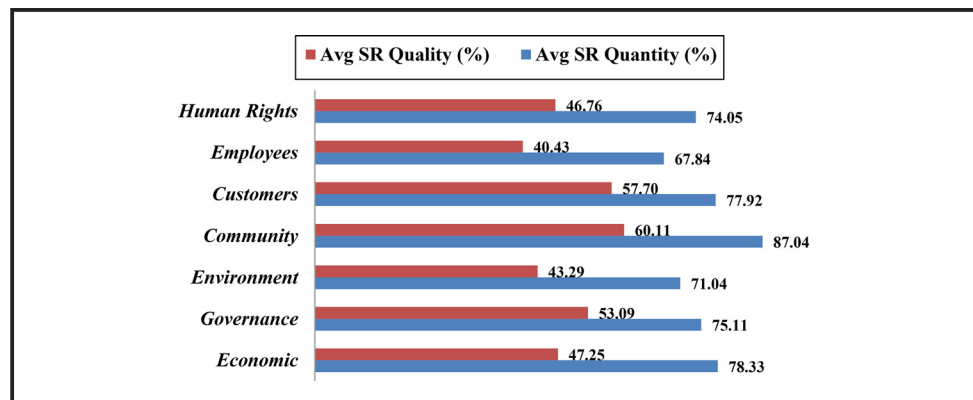


Figure 5 Average SR scores for various sub-categories

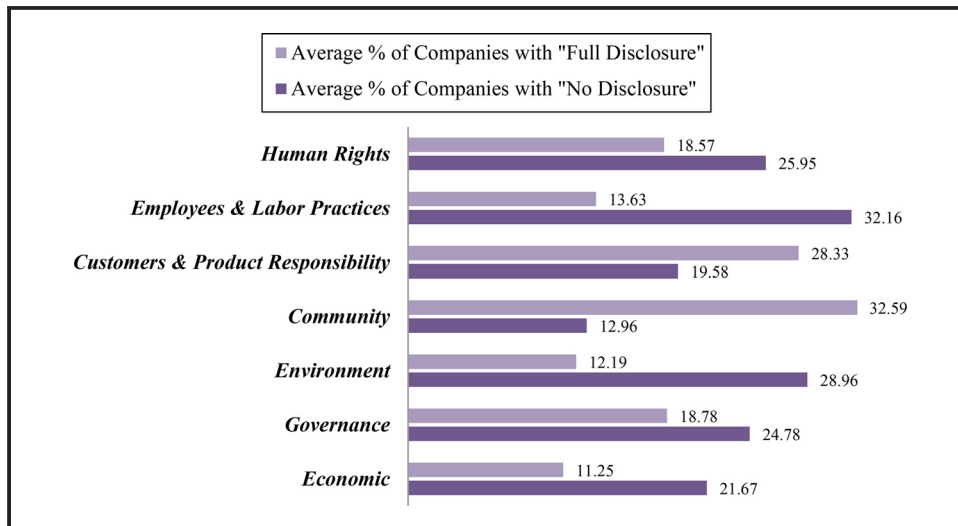
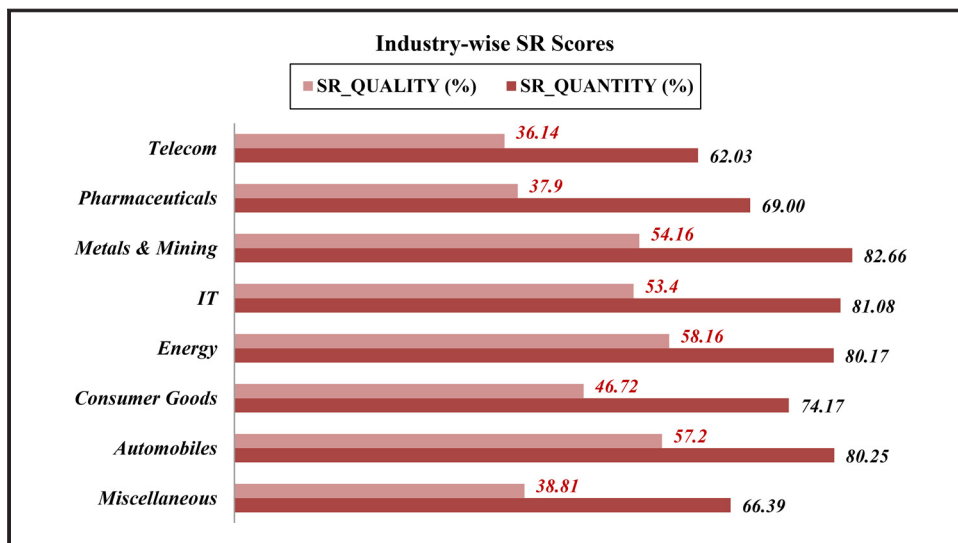


Moreover, both SR quality and SR quantity differ significantly among dimensions ($p = 0.00$). However, eta-squared values for SR quantity and SR quality are 0.115 and 0.141, respectively, suggesting small and medium effect sizes, respectively. Thus, *H2 stands fully supported, but the effect size is not large*. Further, *post-hoc* Tukey's test confirms that community disclosures have the best quality, followed by customers and governance–ethics disclosures.

Subsequently, we deeply analyzed disclosures on various dimensions in terms of no/full disclosure (0 or 5 score). It is found that *one-third companies* have made *full disclosures* on *community items*, providing all the relevant information, followed by customer dimensions (28.33 per cent). However, the least frequency of “full disclosures” belongs to environment (12.19 per cent) and the highest “no disclosures” to employees (32.16 per cent) and environment dimensions (28.96 per cent). In a nutshell, there exists a clear deficiency in the reporting on environment and employee dimensions. Results are exhibited in [Figure 6](#).

Sustainability reporting differences by industry-type

The best SR quality is found in the energy sector, while SR quantity is the highest in the metals and mining sector. The telecom sector turns out the *worst* in both SR quantity and quality, as is demonstrated in [Figure 7](#).

Figure 6 Dimension-wise frequency of full/no disclosure**Figure 7** Average SR scores by industry-type

Besides, a significant difference is found in the SR quantity and quality scores among industries ($p = 0.00$). Eta-squared values are 0.323 and 0.312, respectively, indicating a large effect of industry-type on SR. Thus, $H3$ is fully supported. Further, Tukey's test reveals that among all combinations, only "miscellaneous" sector differs significantly from the energy sector in terms of SR quality ($p = 0.012$) and from metals-mining sector in terms of SR quantity ($p = 0.04$).

Further, 26 companies belong to the ESI. Results suggest that SR quantity and SR quality of ESI companies are significantly higher than other companies ($p < 0.05$), providing support for $H4$. However, eta-squared values indicate *small effect size* as less than 10 per cent variation in SR scores is attributed to the ESI membership, as is shown in Table IV.

Table IV	SR scores of ESI <i>vis-Á-vis</i> non-ESI companies			
<i>Particulars</i>	<i>N</i>	<i>Mean</i>	<i>p-value</i>	<i>Eta-squared</i>
<i>SR_quantity</i>				
ESI company	26	78.79	0.027*	0.081
Non-ESI company	34	71.70		
<i>SR_quality</i>				
ESI company	26	53.05	0.039*	0.071
Non-ESI company	34	45.03		
Note: *Sig. @ 5% level				

Sustainability reporting differences by ownership structure

The sample comprises 13 public-sector companies and 47 privately owned companies. Mean values suggest that public-sector companies perform better in SR. However, the *t*-test output in Table V reveals that results are insignificant. Thus, *H5 is not supported*.

Sustainability reporting and economic performance

Results of the correlation-matrix shown in Table VI reveal that both SR quality and SR quantity have a significant positive correlation with firm size. Thus, large-sized firms apparently perform better in SR. This finding is consistent with previous studies (Liu and Anbumozhi, 2009; Pled and Latridis, 2012; Mertens *et al.*, 2012). However, the correlation between SR and profitability is positive although insignificant, which is similar to previous studies such as Yi and Yu (2010) and Buys *et al.* (2011).

Subsequently, we investigated differences in SR using the *t*-test after dividing the entire sample into two groups based on size (large-/small-sized) and profitability (high/low profitability). Findings suggest that large-sized firms have significantly higher SR quantity and quality scores than small firms. Moreover, the *effect size is large*, Cohen's *d* being 0.86 and 0.76, respectively. However, as regards profitability, results are utterly perplexing, with low-profitable companies having higher mean SR quantity and quality scores. Results are presented in Table VII.

Table V SR Differences by ownership structure			
<i>Particulars</i>	<i>N</i>	<i>Mean</i>	<i>p-value</i>
<i>SR_quantity</i>			
Public sector	13	79.27	0.142
Private sector	47	73.53	
<i>SR_quality</i>			
Public sector	13	55.24	0.067
Private sector	47	46.64	

Table VI Pearson's correlation matrix				
<i>Variables</i>	<i>SR quantity</i>	<i>Profitability (ROA)</i>	<i>Size (Log of market cap)</i>	<i>SR quality</i>
SR quantity	1	0.078	0.486**	0.941**
Profitability (ROA)	0.078	1	0.283*	0.064
Size (Log of market cap)	0.486**	0.283*	1	0.471**
SR quality	0.941**	0.064	0.471**	1
Notes: **Sig. @ 1% level; *Sig. @ 5% level				

Table VII Sustainability reporting differences by economic performance

Particulars	N	Mean (%)	p-value
<i>SR and firm size</i>			
SR quantity			
Large-sized	30	79.70	0.002**
Small-sized	30	69.84	
SR Quality			
Large-sized	30	53.85	0.005**
Small-sized	30	43.16	
<i>SR and firm profitability</i>			
SR quantity			
High profitable	30	73.70	0.509
Low profitable	30	75.84	
SR quality			
High profitable	30	47.15	0.490
Low profitable	30	49.85	
Note: **Sig. @ 1% level			

To sum up, both correlation and *t*-test endorse that large firms perform better in SR but with medium effect size. However, regarding profitability, the results are contradictory. Correlation is positive, while *t*-test suggests an inverse association, although both are statistically insignificant. Overall findings *support H6 but not H7*. Thus, SR and profitability relationship remains inconclusive and warrants further investigation.

Conclusions and discussion

This is a leading study providing valuable insights into SR practices in India, post the mandatory BRR requirement. We developed an all-inclusive 80-item index based on several national and international standards for assessing the SR performance of companies. Content analysis reliability is duly established using Krippendorff's alpha test. Of the seven hypotheses, findings support *H1, H2, H3, H4* and *H6* while not supporting *H5* and *H7*. It is observed that only one-third top-listed companies in India publish standalone CSR/SR, with a major bulk having a "below-average" quality.

GAIL is adjudged as the best company, while Sun TV Network secured the last rank. Although SR quantity and quality crudely provide similar rankings, the SR-quality score is substantially lower than the SR-quantity score. Besides, only one-fourth companies issue externally assured sustainability reports. This raises concerns on the reliability of such disclosures. Thus, companies are advised to publish externally verified sustainability reports containing specific and detailed information with quantitative data to enhance transparency and trustworthiness bringing SR at par with financial reporting.

Further, huge differences are observed in the quantity and quality of CSR disclosures against diverse dimensions. Community-based disclosures are mostly emphasized upon by Indian companies, while employee-related items are the least reported. Thus, *companies in India are perhaps driven by legitimacy theory approach* as community/society is more centered upon than other stakeholders in their haste to gain acceptance from public and society at large, and to augment corporate reputation.

The study also discovered significant differences in SR practices by industry-type with energy sector as the best performer. Also, environmentally sensitive companies are observed making more CSR/sustainability disclosures plausibly to improve their rapport with stakeholders. This finding is consistent with [Patten \(1992\)](#), [Liu and Anbumozhi \(2009\)](#). Telecom companies, being the worst performers, need to toil hard to improve their CSR/sustainability

communications. Further, public-sector companies in India are superior in SR but insignificantly. The positive association so observed between SR and firm size supports the political-cost argument, although effect size is moderate. However, conflicting inconclusive results are observed for profitability indicator. This might be owing to limited sample size and time span of analysis. Also, as CSR/SR regulations in India are relatively a recent phenomenon, the true impact can be assessed only once the gestation phase gets over.

Implications, recommendations and scope for further research

The present study has several implications for corporates, practitioners, policymakers and stakeholders. Low ranking companies need to radically mend their CSR/sustainability communication strategies to meet regulatory and stakeholder demands. Besides, our findings suggest that currently companies in India are driven by regulatory pressure and legitimacy approach to SR. The companies should rather follow stakeholder theory addressing the needs of all stakeholders equitably by making appropriate disclosures on all dimensions, especially with employee-related disclosures being the worst of all. Also, suitable amendments should be made in GRI guidelines and SEBI-BRR framework to avoid patchy responses in the form of yes/no and to encourage specific quantifiable disclosures by companies. Also, a lack of uniformity in SR formats and means adopted by companies calls for a universal standard on SR on similar lines as International Financial Reporting Standards (IFRS), together with mandatory external-assurance, to bring uniformity and reliability in sustainability reporting (Mertens *et al.*, 2012).

It is worth mentioning here that the results of the present study should be interpreted with caution, as it is purely based on self-reported disclosures made by companies in the public domain. They may/may not convey the real performance of the company. The scope of the present study was limited to assessing the frequency and quality of CSR/sustainability disclosures; however, determining their authenticity was beyond the scope of this study. Future researchers should investigate the relationship of SR with profitability and other CFP indicators by performing a longitudinal study with a larger sample of top 500 listed companies. There is also a need to explore integrated reporting, a path ahead of SR, which is still a nascent phenomenon in India. Furthermore, a cross-cultural study across several countries may provide vital acumens on CSR/SR issues.

References

- Abbott, W.F. and Monsen, R.J. (1979), "On the measurement of corporate social responsibility: self-reported disclosures as a method of measuring corporate social involvement", *Academy of Management Journal*, Vol. 22 No. 3, pp. 501-515.
- Ameer, R. and Othman, R. (2012), "Sustainability practices and corporate financial performance: a study based on the top global corporations", *Journal of Business Ethics*, Vol. 108 No. 1, pp. 61-79.
- Barnett, M.L. (2007), "Stakeholder influence capacity and the variability of financial returns to corporate social responsibility", *Academy of Management Review*, Vol. 32 No. 3, pp. 794-816.
- Bayoud, N.S., Kavanagh, M. and Slaughter, G. (2012), "An empirical study of the relationship between corporate social responsibility disclosure and organizational performance: evidence from Libya", *International Journal of Management and Marketing Research*, Vol. 5 No. 3, pp. 69-82.
- Bowen, H.R. (1953), *Social Responsibilities of the Businessman*, Harper and Row, New York, NY.
- Burhan, A.H.N. and Rahmanti, W. (2012), "The impact of sustainability reporting on company performance", *Journal of Economics, Business, and Accountancy/Ventura*, Vol. 15 No. 2, pp. 257-272.
- Buys, P., Oberholzer, M. and Andrikopoulos, P. (2011), "An investigation of the economic performance of sustainability reporting companies versus non-reporting companies: a South African perspective", *Journal of Social Sciences*, Vol. 29 No. 2, pp. 151-158.

- Campbell, D., Craven, B. and Shrives, P. (2003), "Voluntary social reporting in three FTSE sectors: a comment on perception and legitimacy", *Accounting, Auditing & Accountability Journal*, Vol. 16 No. 4, pp. 558-581.
- Carroll, A.B. (1991), "The pyramid of corporate social responsibility: toward the moral management of organizational stakeholders", *Business Horizons*, Vol. 34 No. 4, pp. 39-48.
- Choi, J.S. (1999), "An investigation of the initial voluntary environmental disclosures made in Korean semi-annual financial reports", *Pacific Accounting Review*, Vol. 11 No. 1, p. 73.
- Clarkson, M.B.E. (1995), "A stakeholder framework for analyzing and evaluating corporate social performance", *Academy of Management Review*, Vol. 20 No. 1, pp. 92-117.
- Cormier, D. and Magnan, M. (2003), "Environmental reporting management: a continental European perspective", *Journal of Accounting and Public Policy*, Vol. 22 No. 1, pp. 43-62.
- Daub, C.H. (2007), "Assessing the quality of sustainability reporting: an alternative methodological approach", *Journal of Cleaner Production*, Vol. 15 No. 1, pp. 75-85.
- Deegan, C. (2002), "The legitimacy effect of social and environmental disclosure: a theoretical foundation", *Accounting, Auditing and Accountability Journal*, Vol. 4 No. 1, pp. 32-38.
- Dienes, D., Sassen, R. and Fischer, J. (2016), "What are the drivers of sustainability reporting? A systematic review", *Sustainability Accounting, Management and Policy Journal*, Vol. 7 No. 2, pp. 154-189.
- Faisal, F., Tower, G. and Rusmin, R. (2012), "Legitimising corporate sustainability reporting throughout the world", *Australasian Accounting Business & Finance Journal*, Vol. 6 No. 2, pp. 19-34.
- Frost, G.R. and Wilmshurst, T.D. (2000), "The adoption of environment-related management accounting: an analysis of corporate environmental sensitivity", *Accounting Forum*, Vol. 24 No. 4, pp. 344-365.
- Global Reporting Initiative (GRI) (2011), *G3.1 Sustainability Reporting Guidelines*, Global Reporting Initiative, Amsterdam.
- Guidry, R.P. and Patten, D.M. (2010), "Market reactions to the first-time issuance of corporate sustainability reports: evidence that quality matters", *Sustainability Accounting, Management and Policy Journal*, Vol. 1 No. 1, pp. 33-50.
- Hayes, A.F. and Krippendorff, K. (2007), "Answering the call for a standard reliability measure for coding data", *Communication Methods and Measures*, Vol. 1 No. 1, pp. 77-89.
- Jain, R. and Winner, L.H. (2016), "CSR and sustainability reporting practices of top companies in India", *Corporate Communications: An International Journal*, Vol. 21 No. 1, pp. 36-55.
- Kapoor, S. and Sandhu, H.S. (2010), "Does it pay to be socially responsible? An empirical examination of impact of corporate social responsibility on financial performance", *Global Business Review*, Vol. 11 No. 2, pp. 185-208.
- Khavesh, A., Nikhasemi, S.R., Haque, A. and Yousefi, A. (2012), "Voluntary sustainability disclosure, revenue, and shareholders wealth - a perspective from Singaporean companies", *Business Management Dynamics*, Vol. 1 No. 9, pp. 6-12.
- Kolk, A., van der Veen, M., Wennink, D. and Hay, K. (2002), *KPMG International Survey of Corporate Sustainability Reporting 2002*, KPMG Global Sustainability Services, Netherlands.
- KPMG (2015), "Currents of change: the KPMG survey of corporate responsibility reporting 2015", Publication number: 132962, Haymarket Network Limited, Netherlands, pp. 1-45.
- Laskar, N. and Maji, S.G. (2016), "Corporate sustainability reporting practices in India: myth or reality?", *Social Responsibility Journal*, Vol. 12 No. 4, pp. 625-641.
- Liu, X. and Anbumozhi, V. (2009), "Determinant factors of corporate environmental information disclosure: an empirical study of Chinese listed companies", *Journal of Cleaner Production*, Vol. 17 No. 6, pp. 593-600.
- Manescu, C. (2011), "Stock returns in relation to environmental, social and governance performance: mispricing or compensation for risk?", *Sustainable Development*, Vol. 19 No. 2, pp. 95-118.
- Mertens, G. Maas, K. Strootman, R. and Meliefste, S. (2012), "KPIs and sustainability performance: an empirical analysis concerning the use and development of KPIs on sustainability performance reporting for the largest stock listed companies in The Netherlands", Shareholder Support, Molenberglaan Heerlen, pp. 1-59, available at: www.europroxy.com/Divers/20120315%20ESG%20KPI%20study_final%20version.pdf (accessed 5 November 2014).

- Moneva, J. and Cuellar, B. (2009), "The value relevance of financial and non-financial environment reporting", *Environmental and Resource Economics*, Vol. 44 No. 3, pp. 441-456.
- Morhardt, J.E., Baird, S. and Freeman, K. (2002), "Scoring corporate environmental and sustainability reports using GRI 2000, ISO 14031 and other criteria", *Corporate Social Responsibility and Environmental Management*, Vol. 9 No. 4, pp. 215-233.
- Murray, A., Sinclair, D., Power, D. and Gray, R. (2006), "Do financial markets care about social and environmental disclosure?: Further evidence and exploration from the UK", *Accounting, Auditing & Accountability Journal*, Vol. 19 No. 2, pp. 228-255.
- Neu, D., Warsame, H. and Pedwell, K. (1998), "Managing public impressions: environmental disclosures in annual reports", *Accounting, Organizations and Society*, Vol. 23 No. 3, pp. 265-282.
- Palmer, H.J. (2012), "Corporate social responsibility and financial performance: does it pay to be good?", CMC Senior Theses, Paper 529, Claremont McKenna College, available at: http://scholarship.claremont.edu/cgi/viewcontent.cgi?article=1502&context=cmc_theses (accessed 10 November 2014).
- Patten, D.M. (1992), "Intra-industry environmental disclosures in response to the Alaskan oil spill: a note on legitimacy theory", *Accounting, Organizations and Society*, Vol. 17 No. 5, pp. 471-475.
- Pled, V. and Latridis, G.E. (2012), "Corporate social responsibility reporting: evidence from environmentally sensitive industries in the USA", *International Review of Accounting, Banking & Finance*, Vol. 4 No. 2, pp. 61-99.
- Plumlee, M., Brown, D. and Marshall, S. (2008), "The impact of voluntary environmental disclosure quality on firm value", Working Paper, University of Utah.
- Reverte, C. (2012), "The impact of better corporate social responsibility disclosure on the cost of equity Capital", *Corporate Social Responsibility and Environmental Management*, Vol. 19 No. 5, pp. 253-272.
- Sahu, M. and Panigraphy, D. (2016), "Corporate social responsibility", *Public Sector VS Private Sector - A Myth or Reality? XVII Annual International Seminar, Delhi School of Professional Studies and Research, Delhi*, pp. 124-140.
- Schmelzer, P. (2013), "Value relevance of corporate social responsibility reports", Master's thesis, Erasmus School of Economics, available at: http://thesis.eur.nl/pub/14701/MA369-Schmelzer_320064.pdf (accessed 10 March 2015).
- SEBI (2012), *Business Responsibility Reports*, (CIR/CFD/DIL/8/2012), available at: www.sebi.gov.in/cms/sebi_data/attachdocs/1344915990072.pdf (accessed 10 April 2015).
- Setyorini, C.T. and Ishak, Z. (2012), "Corporate social and environmental disclosure: a positive accounting theory view point", *International Journal of Business and Social Science*, Vol. 3 No. 9, pp. 152-164.
- Udayasankar, K. (2008), "Corporate social responsibility and firm size", *Journal of Business Ethics*, Vol. 83 No. 2, pp. 167-175.
- Vormedal, I. and Ruud, A. (2009), "Sustainability reporting in Norway – an assessment of performance in the context of legal demands and socio-political drivers", *Business Strategy and the Environment*, Vol. 18 No. 4, pp. 207-222.
- Yang, F.J., Lin, C.W. and Chang, Y.N. (2010), "The linkage between corporate social performance and corporate financial performance", *African Journal of Business Management*, Vol. 4 No. 4, pp. 406-413.
- Yi, T.F. and Yu, C.K. (2010), "Research on sustainability reporting in Hong Kong", Project for the Degree of Bachelor of Business Administration (Honours), Hong Kong Baptist University.
- Young, S. and Marais, M. (2012), "A multi-level perspective of CSR reporting: the implications of national institutions and industry risk characteristics", *Corporate Governance: An International Review*, Vol. 20 No. 5, pp. 432-450.

Further reading

- Dominguez, A.A. (2011), "The impact of human resource disclosure on corporate image", *Journal of Human Resource Costing & Accounting*, Vol. 15 No. 4, pp. 279-298.

Table A1 sustainability reporting index

Aspects	Sustainability disclosure items
<i>I. Economic indicators</i>	
A. Economic performance	Direct economic value generated & distributed (including revenues, operating costs, employee compensation, community investments/donations, retained earnings, payments to capital providers and governments) Financial implications and risks/opportunities for organization's activities owing to climate change Coverage of organization's defined benefit plan obligations, including defined contribution plans, percentage of salary contributed by employer/employee and retirement benefits Total monetary value of financial assistance received by organization from government during reporting period Total spending on CSR in amount and percentage of net profits, list of CSR activities with amount spent on each activity
B. Market presence	Ratio of standard entry-level wage to local minimum wage by gender Procedures for local hiring and proportion of senior management hired from local community at significant locations of operation
C. Procurement practices	Proportion of spending on locally based suppliers or percentage of products/services procured locally. Steps taken to improve capability of local/small vendors
8 Items	
<i>II. Governance and ethics indicators</i>	
A. Ethics, transparency and accountability	Organization's values/code of conduct/ethics, policy on bribery/corruption. Whether and how policy extends to suppliers, contractors and joint ventures. If no, explain why Mechanisms for reporting and addressing misconduct/illegal matters. Total number and percentage of governance body members and employees trained on ethics/anti-corruption Total number and percentage of operations/business units assessed for risks related to corruption/unethical practices and significant risks identified Number and nature of incidents of unethical practices/corruption, number of complaints resolved and actions taken
B. Organizational governance	Board committee (s)/executive-level position (s) responsible for sustainability-related decisions and their composition by gender, independence, etc. Whether the Chair of highest governance body is an executive officer (If so, his/her function within organization's management and reasons for this arrangement) Process and criteria used for nominating and selecting highest governance body/committee members Statement from Chairperson/CEO about relevance of sustainability to the organization and strategy for addressing sustainability Process to evaluate highest governance body's performance, particularly regarding sustainability matters and actions taken. Whether such evaluation is independent/self-assessment and its frequency Highest governance body's/senior executives' role in reviewing the implementation of CSR/sustainability policy and its frequency Remuneration policies for highest governance body and senior executives and linkage between remuneration and organization's performance (including social-environmental performance) Ratio of annual total compensation for organization's highest-paid individual to median annual total compensation for all employees, excluding highest-paid individual (pay-gap) Organization's policy and current practice regarding external assurance of sustainability report. If not, report the scope and basis of any external assurance provided. Also, report the relationship between organization and assurance providers
C. Stakeholder engagement	List of stakeholder groups engaged by organization. Identify the disadvantaged/vulnerable/marginalized stakeholders and the basis for identification of stakeholders so engaged Organization's approach to stakeholder engagement, including frequency of engagement by type and stakeholder group Key topics/concerns raised through stakeholder engagement and organization's response to them

(continued)

Table A1

Aspects	Sustainability disclosure items
15 Items	
<i>III. Environmental indicators</i>	
A. Water	Total volume of water consumed/withdrawn from various sources and the water sources significantly affected by organization Water conservation initiatives, reduction in water consumption and volume and percentage of water recycled/reused
B. Energy	Direct and indirect energy consumption from various renewable and non-renewable sources Energy conservation initiatives and total reduction in energy consumption
C. Effluents and waste	Total weight of waste by type (hazardous/not), disposal method and percentage of recycling of products/waste Total number, volume and impacts of recorded significant spills on soil/water surfaces (e.g. oil/fuel/chemical spills) Total wastewater and effluents discharged by quality (treated/not) and destination; and water bodies/habitats significantly affected by discharges/run-off
D. Biodiversity	Operational sites in/around protected areas and areas of high biodiversity value and significant impacts on biodiversity in protected areas Measures for managing impacts on biodiversity
E. Emissions	Direct and indirect GHG, NOx, SOx, CO ₂ and ozone depleting substances emissions Amount of GHG, carbon and other emissions reduced and initiatives taken to reduce emissions
F. Materials, suppliers and transportation	Total weight/volume of renewable and non-renewable materials and percentage of recycled input materials used Significant environmental impacts of transporting products/materials/workforce and how such impacts are mitigated Suppliers' Environmental Impact Assessment; percentage of new suppliers/contractors screened using environmental criteria; significant negative environmental impacts identified and actions taken
G. Others	Number of grievances about environmental impacts filed and resolved through formal grievance mechanism. Are emissions/waste generated by company within permissible limits? Any show cause/legal notices received from Pollution Control Board pending at year end. Fines/sanctions for non-compliance with environmental laws/regulations Total amount of environmental protection expenditures and investments by type
16 Items	
<i>IV. Community indicators</i>	
A. Culture, education and inclusive growth	Protect national heritage, art and culture. Promote cultural activities, traditional arts/handicrafts and sports (especially rural sports) Promote literacy/education Promote women empowerment Inclusive growth and equitable development – employability enhancement activities, social infrastructure investments, rural/slum area development, etc.
B. Community health	Promote community health and sanitation
C. Responsible public policy advocacy	List major memberships of trade/industry associations/chambers and broad areas advocated/lobbied through above platforms Total monetary value of financial and in-kind political contributions made directly/indirectly by country and recipient/beneficiary
D. Others	Impact assessments and outcomes of community engagement/development initiatives. Steps taken to ensure successful adoption of initiatives by the community Number of grievances about impacts on society/community filed and resolved through formal grievance mechanism
9 Items	
<i>V. Customers and product responsibility indicators</i>	
A. Sustainable consumption	List up to three products/services whose designs incorporate social/environmental concerns Policy on products' life-cycle sustainability assessment and sustainable value chain
B. Customer value	Provide value to customers (e.g. customer health and safety, good quality, reasonable prices and after sales support) Report cases of sale of banned/disputed/hazardous products
C. Responsible marketing communications	Responsible marketing communications

(continued)

Table A1

Aspects	Sustainability disclosure items
D. Customer privacy	Use adequate safeguards to protect consumer privacy and personal data. Report cases of breach
E. Customer feedback and grievances	Customer grievance handling mechanism and customer satisfaction survey. Report key results of surveys. Provide details of customer complaints received and percentage of complaints/consumer cases pending at year end Number of cases filed against company during last five years regarding irresponsible labeling/ advertising/unfair trade practices/anti-competitive behavior and those pending at year end. Provide details of cases and outcomes
8 Items	
<i>VI. Employees and labor practices indicators</i>	
A. Employment	Total number and percentage of employees by employment-type, gender, age, disabilities, etc. Total number and rate of new employees hired and employee turnover by age group and gender Benefits provided only to full-time employees and not to temporary/part-time employees Amount of minimum monthly wage paid to skilled and unskilled employees Ratio of basic salary and remuneration of women to men by employee category
B. Labor-management relations	Recognized trade unions/employee associations; percentage of permanent employees who are members of such associations; and percentage of total employees covered by collective bargaining agreements Reasonable/minimum notice period provided to employees about operational changes. Whether period is specified in collective bargaining agreements
C. Occupational health and safety (OHS)	Employee health and safety measures Formal joint management-worker OHS committee and percentage of workforce represented in committee. Report the extent to which health/safety topics are covered in formal trade union agreements Report percentage of employees given safety training by employee category Report injury rate, occupational diseases rate, lost day rate, absentee rate and work-related fatalities for total workforce by gender
D. Training and development	Average hours of training per year per employee and percentage of employees trained by employee category Describe programs for training, skills management and career endings management/retirement planning organized during the reporting period Performance review mechanism and percentage of employees regularly reviewed by gender and employee category
E. Conditions of work	Sexual harassment prevention policy and committee. Number of complaints on sexual harassment received and pending at year end Work-life balance facilities, especially for women and vulnerable employees. Also, report return to work and retention rates of employees taken parental leave by gender
F. Supplier assessment	Suppliers'/Contractors' Labor Practices Assessment and percentage of new suppliers/contractors so screened. Report significant negative impacts identified and actions taken
G. Labor grievance redressal mechanism	Number of grievances about labor practices filed and resolved through formal grievance mechanism
17 Items	
<i>VII. Human rights indicators</i>	
A. Human rights assessment and training	Total hours of employee training on human rights; and percentage of employees and security personnel trained Avoid complicity in human rights abuse. Report total number and percentage of investment agreements/ suppliers/contractors/joint ventures subjected to human rights assessment. Report significant negative impacts identified and actions taken
B. Non-discrimination and equal opportunities	Policy on non-discrimination and equal opportunities for all. Number of incidents of discriminatory employment corrective actions taken. Report-related significant fines/sanctions
C. Freedom of association and collective bargaining	Policy on freedom of expression/association and collective bargaining. Report operations/suppliers where such rights may be violated/risked and actions taken
D. Child labor and forced or compulsory labor	Policy on child/forced labor. Number of complaints on child/forced/involuntary labor received and pending at the end of year. Also, report measures taken to contribute to elimination of child/forced labor
E. Indigenous rights	Respect rights of indigenous people living in communities surrounding company's operations. Number of incidents of violations involving rights of indigenous people and actions taken such as rehabilitation
F. Human rights grievance redressal mechanism	Number of complaints/grievances about human rights received and resolved through formal grievance mechanism
7 Items	
Total: 80 Items	

Appendix 2

Table All Sustainability reporting scores and ranks of sample companies

S. No.	Company	No. of items disclosed	SR quantity score	SR quality score	Rank (SR quality)	Rank (SR quantity)
1	GAIL Ltd.	74	92.5	79.25	1	4
2	Cairn India Ltd.	69	86.25	73.5	2	14
3	Tata Steel Ltd.	75	93.75	73.25	3	2
4	Tata Motors Ltd.	74	92.5	73.25	3	4
5	Tata Power Company Ltd.	76	95	73	5	1
6	BPCL	70	87.5	71	6	12
7	RIL	74	92.5	70.5	7	4
8	IOC Ltd.	75	93.75	70.25	8	2
9	Mahindra & Mahindra Ltd.	72	90	70	9	10
10	TCS Ltd.	74	92.5	69.37	10	4
11	Infosys Ltd.	73	91.25	68.61	11	8
12	ITC Ltd.	73	91.25	68.25	12	8
13	Maruti Suzuki India Ltd.	70	87.5	67	13	12
14	L & T Limited	71	88.75	63	14	11
15	OIL	67	83.75	61.75	15	18
16	Coal India Ltd.	64	80	58.75	16	22
17	NTPC Limited	64	80	58	17	22
18	JSW Steel Ltd.	68	85	57.75	18	16
19	Dabur India Ltd.	58	72.5	56.25	19	31
20	Sesa Sterlite Ltd.	69	86.25	53.5	20	14
21	Colgate India Ltd.	68	85	52.75	21	16
22	ONGC Ltd.	59	73.75	52	22	30
23	Cummins India Ltd.	60	75	51.5	23	28
24	Titan Company Limited	65	81.25	51.25	24	19
25	NMDC Ltd.	65	81.25	48.75	25	19
26	HUL	64	80	48.5	26	22
27	Hindustan Zinc Limited	65	81.25	48.25	27	19
28	Jindal Steel & Power Ltd.	61	76.25	48	28	27
29	Wipro Ltd.	63	78.75	46.75	29	25
30	SAIL	62	77.5	45	30	26
31	BHEL	57	71.25	45	30	35
32	Power Grid Corporation of India Ltd.	60	75	44.81	32	28
33	Dr. Reddy's Laboratories Ltd.	58	72.5	43.25	33	31
34	Container Corporation of India Ltd.	53	66.25	43.04	34	42
35	NHPC Ltd.	53	66.25	41	35	42
36	Bajaj Auto Ltd.	52	65	40.25	36	48
37	Grasim Industries Ltd.	55	68.75	39.5	37	36
38	Godrej Consumer Products Ltd.	58	72.5	39.25	38	31
39	Cipla Ltd.	53	66.25	39.25	38	42
40	Idea Cellular Ltd.	55	68.75	38.73	40	36
41	DLF Ltd.	53	66.25	37.25	41	42
42	Cadila Healthcare Ltd.	55	68.75	37.25	41	36
43	Aditya Birla Nuvo Ltd.	54	67.5	37.25	41	40
44	Bharti Airtel Ltd.	44	55	37.22	44	56
45	United Breweries Ltd.	54	67.5	37	45	40
46	Asian Paints Ltd.	53	66.25	37	45	42
47	UltraTech Cement Ltd.	55	68.75	36	47	36
48	Hero Motocorp Ltd.	53	66.25	36	47	42
49	Lupin Ltd.	52	65	35.25	49	48
50	Reliance Communications Ltd.	49	61.25	34.68	50	51

(continued)

Table All

<i>S. No.</i>	<i>Company</i>	<i>No. of items disclosed</i>	<i>SR quantity score</i>	<i>SR quality score</i>	<i>Rank (SR quality)</i>	<i>Rank (SR quantity)</i>
51	Sun Pharmaceuticals Industries Ltd.	58	72.5	34.5	51	31
52	Adani Enterprises Ltd.	47	58.75	34.5	51	54
53	Adani Power Ltd.	50	62.5	34	53	50
54	Bharti Infratel Ltd.	48	60	33.92	54	52
55	Adani Ports and SEZ Ltd.	48	60	32.91	55	52
56	Marico Industries Ltd.	41	51.25	30.25	56	59
57	OFSS Ltd.	47	58.75	28.86	57	54
58	Reliance Power Ltd.	42	52.5	27.5	58	58
59	Zee Entertainment Enterprises Ltd.	44	55	26.58	59	56
60	Sun TV Network Ltd.	38	47.5	19.24	60	60

Corresponding author

Priyanka Aggarwal can be contacted at: priyanka_aggarwal6889@yahoo.in

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgroupublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com